

Question Number	Answer	Additional Guidance	Mark
1a	An answer that includes the following: • convert starch to maltose (1)	Allow convert amylose / amylopectin Allow breakdown / digest / hydrolyse Allow convert to glucose	1

Question Number	Answer	Additional Guidance	Mark
1b	A description that includes six of the following points: • use of aseptic technique (1) • {cut seed in half / to separate endosperm from embryo} (1) • soak both halves in gibberellin (1) • use of control (from grains) not soaked in gibberellin (1) • place (grain halves cut side down) on starch agar plate (1) • incubate for a stated time (range 24-48 hours) / at a {stated / fixed} temperature (1) • add iodine (to plate after incubation) (1) • description of how results can be compared (1)	Allow wash with sterile water / sodium hypochlorite also in preparing plate Allow soak in water Do not allow waterbath	6

Question Number	Answer	Additional Guidance	Mark
1c	A description that includes three of the following points: • gibberellin is a transcription factor (1) • increasing the transcription (of amylase mRNA) (1) • (resulting in) translation of amylase (1)	Ignore genes / cAMP Allow starts / stimulates / enables transcription Allow enzyme / protein	3

(Total for question 1 = 10 marks)